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3 Mitosis and cell division are involved in growth and repair in plants and animals.

- (a) (i) Name the stages of mitosis when chromosomes line up across the centre of the cell and when chromosomes separate.

(1)

Chromosomes line up

Chromosomes separate

- (ii) Cell division takes place after mitosis.

Compare and contrast cell division in animal and plant cells.

(3)

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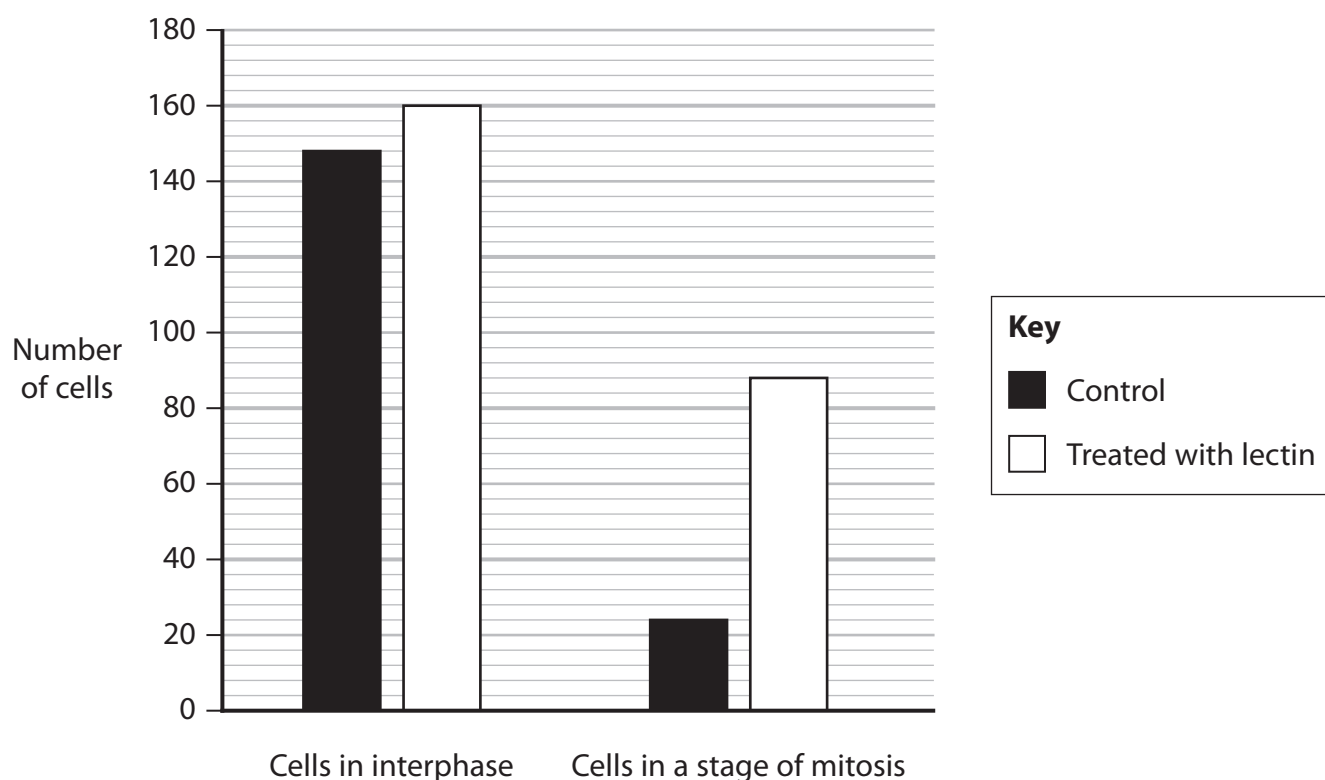
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- (b) The hypothesis that the addition of the protein lectin stimulates mitosis in onion cells was investigated. The graph shows the results of this investigation.



- (i) State the independent variable in this investigation.

(1)

- (ii) Describe a suitable and safe procedure to obtain all the data shown in the graph.

(5)



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(iii) Draw a table to show these results.

(3)



- (iv) In control group cells 14.0% are in mitosis. Calculate the percentage of cells in mitosis in those treated with lectin.

(2)

Answer %

- (v) Comment on the conclusion that these results show that lectin does stimulate mitosis.

Use the information in the graph and your answer to (b)(iv) to support your answer.

(2)

- (vi) Justify the selection of a named statistical test to be performed on these data.

(2)

Statistical test

Reason

(Total for Question 3 = 19 marks)

TOTAL FOR PAPER = 50 MARKS

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